

IILM UNIVERSITY, GREATER NOIDA		LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET		
EXPERIMENT TITLE: To determine the resistance per unit length of Carey Foster's bridge wire and specific resistance of the material of the given wire using Carey Foster's bridge.		
EXPERIMENT NO. 05	DEPT. : Physics	ISSUE DATE: 12 August, 2026
LABORATORY: 26SOSP101 Semiconductor and Quantum Physics Lab	SEMESTER: I, II	TOTAL PAGE: 4
AIM: To determine the resistance per unit length of Carey Foster's bridge wire and specific resistance of the material of the given wire using Carey Foster's bridge. APPARATUS: Carey Foster's bridge, rheostat, Battery eliminator, copper strip, galvanometer, plug key, given wire whose specific resistance is to be determined, resistance box, connecting wires and screw gauge. THEORY: The resistance per unit length of Carey's Foster's bridge wire (ρ) is related to the difference between two nearly equal resistances connected in the outer gaps of a Carey Foster's bridge as, $X - Y = \rho (L_2 - L_1) \quad (1)$ where X = Resistance connected, Y = Resistance of copper strip connected to the right gap ρ = Resistance per unit length of the bridge wire. L_1, L_2 = length of balance point on the bridge wire measured from the left reference zero before and after interchanging the X and Y resistances. The specific resistance (k), of material of a given wire can be calculated from the formula: $k = \frac{R \pi r^2}{L} \quad (2)$ where, R = resistance of the given wire L = length of the given wire r = radius of the given wire The resistance of the given wire is given as:		

$$R = X - \rho(L'_2 - L'_1)$$

where, L'_1 and L'_2 are the length of balance points measured from left reference zero before and after interchanging the resistance introduced (X) and given wire in place of the copper strip (Y).

DIAGRAM:

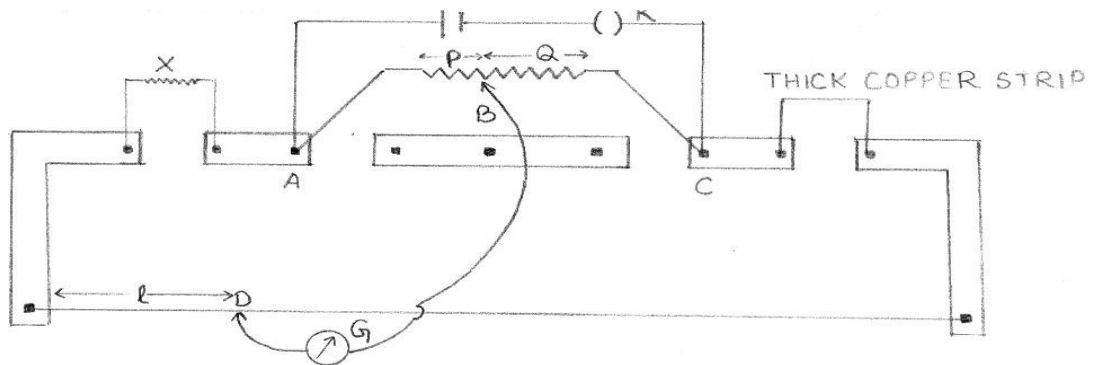


Fig 1: Experimental arrangement of Carey Foster Bridge

PROCEDURE:

(1) Method for Determination of the Resistance per Unit Length of the Bridge Wire

1. The standard resistance is kept in the outer left gap for X and thick copper strip in the outer right gap for Y (fig (1)). The ratio arms P and Q can be obtained by connecting the lower terminals of the rheostat to the points A and C and its upper terminal kept at the middle and connected to the sliding contact to the galvanometer G.
2. The key (K) is closed and the jockey is then made to touch the bridge wire near its ends. The deflections in the galvanometer in the two cases must be in opposite directions. However, if only one-sided deflection is observed then the connections should be rechecked and screws properly tightened, seeing that the ends of the connecting wire are perfectly clean.
3. Introduce some resistance X in the resistance box and by sliding the jockey null point is obtained. The distance of null point L_1 from left reference zero ends is noted.
4. Now interchange the position of resistance box and the copper strip. Again the balance is obtained for the same applied resistance in the box. Let the null point from the left reference

zero be L_2 .

- Change the value of X by 1ohm and obtain different sets of observation as described in above steps.
- Calculate the value of ρ separately by taking $Y=0$ for each set using the formula:

$$\rho = \frac{X}{(L_2 - L_1)} \text{ and obtain the mean value of } \rho.$$

(2) METHOD FOR DETERMINING THE RESISTANCE OF THE GIVEN WIRE: -

The experiment is repeated exactly in the same manner as stated above by replacing the copper strip with the given wire and retaining the same resistance X. The resistance of the given wire R, is then calculated by the value of ρ by,

$$R = X - \rho(L'_2 - L'_1)$$

Take different sets by changing the value of resistance in the resistance box.

OBSERVATIONS: [A] OBSERVATIONS FOR DETERMINATION OF ρ :

S. No.	Standard resistance used (X)	Distance of null point from the left end of bridge wire		$(L_2 - L_1)$ cm.	$\rho = \frac{X}{(L_2 - L_1)}$ $\Omega \text{ cm}^{-1}$
		Resistance X in left gap and copper strip in right gap (L_1 cm).	Resistance X in right gap and copper strip in left gap (L_2 cm).		
1.					
2.					
3.					
4.					

The mean value of ρ =..... $\Omega \text{ cm}^{-1}$

[B] OBSERVATIONS FOR DETERMINATION OF THE RESISTANCE OF THE GIVEN**WIRE:**

S. No.	Standard resistance used (X)	Distance of null point from left end of the bridge wire		$(L'_2 - L'_1)$ cm.	Resistance of the wire $R = X - \rho(L'_2 - L'_1) \Omega$
		Resistance X in the left gap and given wire in the right gap(L'_1 cm).	Resistance in the right gap and given wire in the left gap (L'_2 cm)		
1.					
2.					
3.					
4.					

The mean value of $R = \dots\dots\dots \Omega$

RESULT: - Resistance per unit length of Carey Foster's bridge wire = $\dots\dots\dots \Omega \text{ cm}^{-1}$

Resistance of the material of the given wire = $\dots\dots\dots \Omega$.

PRECAUTION AND SOURCES OF ERROR:

1. A key should be used and it should be closed only while observations are being made.
2. While checking the null point the cell key should be closed first and then the jockey should be made to touch the bridge wire.
3. The jockey should be pressed gently and momentarily. It should not be dragged along the length of wire otherwise it will spoil the bridge wire.
4. The difference between X and Y should not be more than the resistance of the bridge wire.

VIVA QUESTIONS:

1. What is a Carey Foster Bridge?
2. What is the principle of the Carey Foster Bridge?
3. What is the role of bridge wire?
4. What is the function of the galvanometer in this experiment?